

ENGR 3323: Signals and Systems

HW 4_Ch2 Answer Keys

$$\mathbf{1a)} (1 - e^{-2t}) u(t) \quad \mathbf{1b)} (e^{-t} - e^{-2t}) u(t) \quad \mathbf{1c)} \frac{0.83e^{-2t} - \cos(3t - 326^\circ)}{\sqrt{13}}$$

$$\mathbf{d)} (1 - e^{-2t}) \text{ for } 0 < t < 2 \text{ and } 53.6e^{-2t} \text{ for } t > 2 \quad \mathbf{e)} e^{-2t} u(t) \quad \mathbf{f)} e^{-2(t-2)} u(t-2)$$

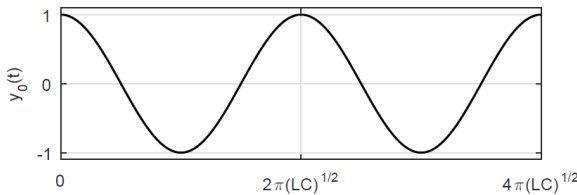
$$\mathbf{2a)} y(-1) = 0; \quad y(0) = 1; \quad y(1) = 1; \quad y(2) = 1; \quad y(3) = 1; \quad y(4) = 1; \quad y(5) = 1; \quad y(6) = 0;$$

2b)

$$y(t) = \begin{cases} 0 & t < -1 \\ t + 1 & -1 \leq t < 0 \\ 1 & 0 \leq t < 5 \\ 6 - t & 5 \leq t < 6 \\ 0 & t \geq 6 \end{cases}$$

$$\mathbf{3a)} \frac{d^2 y}{dt^2} + \frac{1}{LC} y(t) = \frac{1}{LC} x(t) \quad \mathbf{3b)} \lambda_{1,2} = \frac{\pm j}{\sqrt{LC}} \quad \mathbf{3c)} y_0(t) = \cos\left(\frac{t}{\sqrt{LC}}\right)$$

3d)



3e) The zero state response

$$x(t) * h(t) = (0.5 \sin(t) - 0.5 \cos(t) + 0.5e^{-t}) u(t)$$

The total response

$$y(t) = (0.5 \sin(t) + 0.5 \cos(t) + 0.5e^{-t}) u(t)$$

4a) The system is BIBO stable and also asymptotically stable

4b) The system is BIBO stable and marginally stable

4c) The system is BIBO unstable and asymptotically unstable

4d) The system is BIBO unstable and asymptotically unstable

5)

$$\lambda = \frac{-2 \pm \sqrt{4 - 13}}{2} = -1 \pm j\frac{3}{2}.$$

$\cos(\omega t)$ will produce a strong response when $\omega = \pm \frac{3}{2}$ rad/s.